

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5107

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5

Total Marks:50

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

*I J.VISHNU VARDHAN(192324297) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*



Signature of the student:

TECHNICAL PRESENTATION

1. From Caesar to Vigenère: Evolution of Substitution Ciphers”
2. “Substitution vs Transposition: Which Provides Better Security?”
3. “Security Attacks, Threats, Risks and Their Impact on Security Goals”
4. “Steganography vs Cryptography: Hidden Security in the Digital World”
5. “Role of Number Theory in Cryptography: From Euclid to Euler”

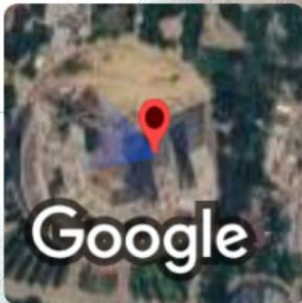
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Presentation Geo-Tag Photo:



GPS Map Camera



Mevalurkuppam, Tamil Nadu,
India 🇮🇳

22g8+cwh, Mevalurkuppam, Tamil Nadu 602105,
India

Lat 13.026496° Long 80.016862°

Thursday, 08/10/2026 02:45 PM GMT +05:30



Steganography vs Cryptography: Hidden Security in the Digital World

Presented by:

J. Vishnu Vardhan



Introduction

Steganography and Cryptography protect digital information.

Cryptography hides the meaning of a message.

Steganography hides the existence of a message.

Both improve data security and privacy.

Used in cybersecurity and secure communication.



Cryptography & Steganography



CRYPTOGRAPHY

- Converts plaintext into unreadable ciphertext.
- Uses encryption and decryption techniques.
- Examples: AES, RSA, Caesar Cipher.
- Protects the content of information.

STEGANOGRAPHY

- Hides secret information inside another file.
- Can use images, audio, video, or text.
- Example: LSB image steganography.
- Protects the existence of information.



Steganography vs Cryptography



Cryptography → Hides message meaning

Steganography → Hides message existence

Cryptography → Produces ciphertext

Steganography → Produces a normal-looking file

Cryptography → Commonly uses encryption keys

Steganography → May use a secret key

Cryptography → Cryptanalysis is used to attack it

Steganography → Steganalysis is used to detect it



Applications & Conclusion



Applications:

- Secure military communication
- Digital watermarking
- Copyright protection
- Confidential communication
- Secure data transfer

Conclusion:

- Cryptography protects the content.
- Steganography hides the presence.
- Combining both can provide stronger security.



THANK YOU!